5.7 Corresponding Parts of Congruent Triangles Are Congruent

Lesson Introduction

Benchmarks	MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Angle, Angle-Angle, and Hypotenuse-Leg. MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures. MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.	Learning Targets	• I understand how to use corresponding parts of congruent triangles in a proof. • I can prove parts of triangles are congruent using triangle congruence and corresponding parts of congruent triangles.

Benchmark Coherence	Building On	Working Toward
	N/A	N/A

Essential Questions	• How can I prove corresponding parts of congruent triangles are congruent using congruent triangles?
Lesson Narrative	In this lesson, students continue the work they did in the prior lesson on proving that two triangles are congruent. For this lesson, students will include the use of corresponding parts of congruent triangles to prove congruence of other parts. Students will work to complete proofs in various formats to first prove triangle congruence, then to prove the congruence of corresponding parts of congruent triangles.

Component Summary				
	5.7.1 Warm-Up (~5 min.)	Medieval brain teaser *(SE p. 262)*	5.7.4 Guided Practice (5-10 min.)	Practice with completing proofs using CPCTC *(SE p. 266)*
	5.7.2 Collaboration (10 min.)	Collaboration on congruency proofs using overlapping triangles *(SE p. 263)*	5.7.5 Your Turn! (5 min.)	Practice with completing proofs using CPCTC *(SE p. 266)*
	5.7.3 Guided Instruction (10-15 min.)	Formal instruction on completing proofs using CPCTC *(SE p. 264)*	5.7.6 Wrap-Up (~5 min.)	Self-reflection on proving triangles congruent *(SE p. 267)*

Resources	Glossary	Materials	Additional Preparation
	<u>Key Terms</u>: • Corresponding Parts of Congruent Triangles are Congruent (CPCTC) <u>Additional Terms</u>: N/A	• (Optional) Colored Markers • (Optional) Patty Paper	• Students may struggle with the proofs in this lesson. Work through each one on your own prior to delivering instruction to work through potential misconceptions in the proofs.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may not realize all the information that is available in a given problem when it’s not explicitly stated.
Students may apply the same marking to multiple sides, causing confusion.
Students may struggle with proving the triangles congruent.
Students may struggle with overlapping triangles.</td><td>Consider having students keep a list of definitions, postulates, and theorems. The list can include common conclusions. For example, if given two lines are parallel, it’s common for congruent angles to be the next statement in a proof.
If time is an issue, consider assigning 5.7.5 Your Turn! for homework.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think-Pair-Share
Structure 5.7.4 Guided Practice as a Think-Pair-Share. Students can work on their own to try to complete the paragraph proof and then pair up with a partner to determine the correct answers by sharing their thinking and approaches with one another.</td><td>MA.K12.MTR.5.1: Patterns and Structure
Students will be working to complete formal proofs that show two triangles are congruent. They will need to logically order the events in such a way that it solves the problem of proving the two triangles are congruent.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider placing students in groups based on their performance so far in this unit. Students will be applying their knowledge from prior lessons through the process of completing a proof. Students who struggled with triangle congruency will likely struggle proving it.
For activities that require you to review proofs (including reviewing the answers), consider using auditory explanations for a proof prior to filling in the proof. Use common language to describe how the triangles could be proven congruent. Then, transition that language into formal geometric language when completing the proof.
The 5.7.5 Your Turn! provides students an opportunity to demonstrate their understanding of triangle congruence using a proof. Provide students with the opportunity to choose to complete a two-column proof or paragraph proof to demonstrate their understanding. Use their work as a starting point of remediation toward completing flowchart proofs.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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5.7.1 Warm-Up: Brain Teaser

Component Overview: *Students will complete a medieval brain teaser. Students may benefit from drawing a diagram of the man's journey before writing a summary.*

5.7.1 Warm-Up: Brain Teaser

Brain teasers like this have been traced back to medieval times. Here's one with a Florida twist.

1. A man has to take a panther, a raccoon, and some strawberries across St. John's river. His rowboat has enough room for the man, plus either the panther or the raccoon or the strawberries. If he takes the strawberries with him, the panther will eat the raccoon. If he takes the panther, the raccoon will eat the strawberries. Only when the man is present are the raccoon and the strawberries safe from their enemies. All the same, the man successfully carries panther, raccoon, and strawberries across the river. How?
   The man takes the raccoon over to the other side. He goes back and gets the strawberries. Once he delivers the strawberries, he also brings the raccoon back to the first side. He leaves the raccoon and takes the panther back to the other side. He returns one more time to bring the raccoon to the side he is trying to get to.

Watch for the pacing on this activity. Students may get lost in the wording of this brain teaser and want additional time to complete. If students are unable to complete this during the allotted time, consider allowing them time at the end of class to complete.

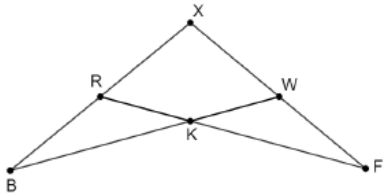
5.7.2 Collaboration: Overlapping Triangles

Component Overview: *Students will work together to complete proofs involving overlapping triangles. For this activity, release students to complete the entire activity with a partner prior to coming back together as a whole class and reviewing the answers together.*

5.7.2 Collaboration: Overlapping Triangles

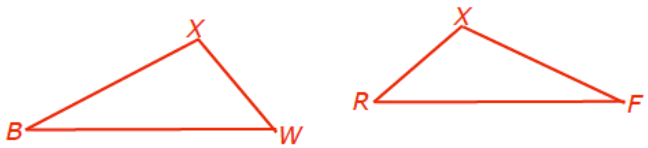
Work with your partner to complete the following.

Polygon $XFKB$ is shown.



1. How many triangles are in polygon $XFKB$?
   4 triangles

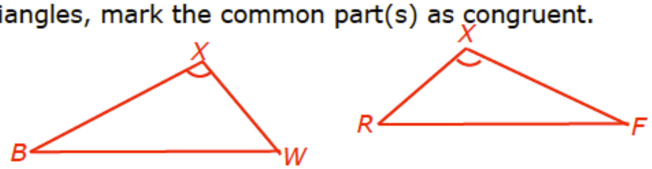
2. Draw ΔXBW and ΔXFR separately.



3. Discuss with your partner if triangles ΔXBW and ΔXFR have any common sides or angles. If so, name the common part(s).
   $\angle X \cong \angle X$

4. Explain why knowing if ΔXBW and ΔXFR have any common sides or angles could be helpful in a proof.
   Answers will vary. Knowing if the triangles have any common sides or angles will identify if there are any congruent parts shown in the figure.

5. In your drawing of ΔXBW and ΔXFR as separate triangles, mark the common part(s) as congruent.



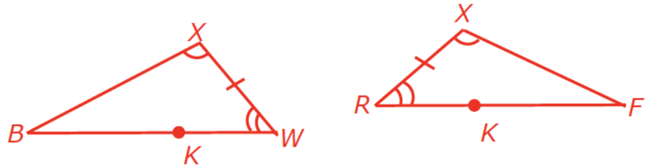
For this activity, allow students to work with their partner to complete all the questions on their own. Consider leveraging this time to work with students who have been struggling with grasping the material in prior lessons.

How can you recognize when the reflexive property of congruence can be used?

5.7.2 Collaboration: Overlapping Triangles (continued)

6. What property justifies that something is equal to itself?
   Reflexive Property of Congruence

7. Mark the given information in your diagram.
   Given: $\overline{RX} \cong \overline{WX}$, $\angle XRK \cong \angle XWK$



8. What can you conclude about ΔXBW and ΔXFR from the information you marked on your drawing?
   $\Delta XBW \cong \Delta XFR$ by Angle-Side-Angle Congruence Postulate.

9. Complete the flowchart proof for polygon $XFKB$.
   Given: $\overline{RX} \cong \overline{WX}$, $\angle XRK \cong \angle XWK$
   Prove: $\angle XBK \cong \angle XFK$

$\overline{RX} \cong \overline{WX}$
Given

$\angle XRK \cong \angle XWK$
Given

$\angle X \cong \angle X$
Reflexive Property of Congruence

$\Delta XBW \cong \Delta XFR$
Angle-Side-Angle Congruence Theorem

$\angle XBK \cong \angle XFK$
CPCTC

10. Check your proof with another group and if necessary, make adjustments.

In question 7, students may wonder where point K is in their drawings. Direct students back to the original image to locate K for determining the correct angle.

Consider adding a task in between questions 7 and 8 for this activity. Have student pairs locate an additional student pair to verify their markings to ensure there isn't something they missed.

For a challenge, consider adjusting question 10. Have students trade their proof with another group and be prepared to explain their thinking to an additional group. Each student pair will see two other pairs in this process.

5.7.3 Guided Instruction: Proofs Using CPCTC

Component Overview: *Students will receive guided instruction on completing proofs using CPCTC. For this activity, guide students through all parts of the question and allow students to work with a partner when it's indicated.*

5.7.3 Guided Instruction: Proofs Using CPCTC

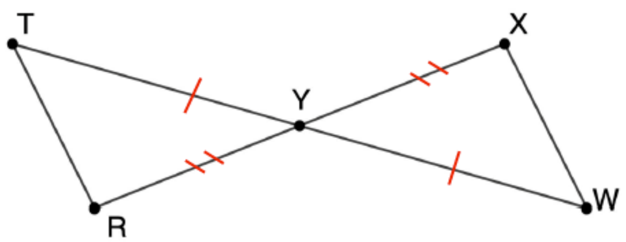
Once two triangles have been proven to be congruent, then CPCTC states that any other corresponding parts of the triangles are also congruent.

Corresponding Parts of Congruent Triangles are Congruent (CPCTC)

If two triangles are congruent, then all corresponding parts of the triangles are congruent.

In the flowchart proof, proving that $\angle XBK \cong \angle XFK$ first required proving that ΔXBW and ΔXFR are congruent.

1. Triangles TYR and XYW are shown.



a. Mark the diagram to show the following given information.
Given: Y is the midpoint of $\overline{WT}$. Y is the midpoint of $\overline{RX}$.
Marked on diagram.

b. What does "Y is the midpoint of $\overline{WT}$" allow you to conclude about the diagram?
"Y is a midpoint of $\overline{WT}$" allows me to conclude that $\overline{TY} \cong \overline{YW}$.

Students have already learned about CPCTC in prior units.

Questions to consider:

- *How can you determine where your markings should go?*
- *What process would you go through to look for additional information that is not listed as "given"?*
- *How can there be additional information that you can mark without it being provided explicitly in the "given"?*

c. What does "Y is the midpoint of $\overline{RX}$" allow you to conclude about the diagram?
"Y is a midpoint of $\overline{RX}$" allows me to conclude that $\overline{RY} \cong \overline{YX}$.

d. Discuss with your partner if there is any other information about the two triangles that can be determined from the diagram. Write a summary of your discussion.
Answers will vary. $\angle XYW \cong \angle TYR$ because of the Vertical Angles Theorem.

e. Work with your partner to complete the paragraph proof.

Given: Y is the midpoint of $\overline{WT}$. Y is the midpoint of $\overline{RX}$.
Prove: $\overline{TR} \cong \overline{XW}$

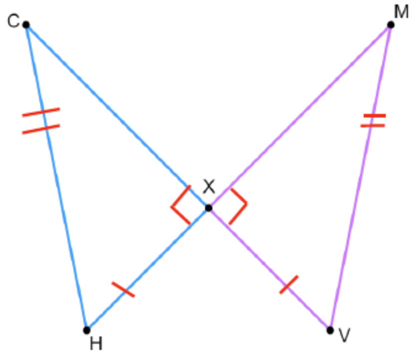
It is given that Y is the midpoint of $\overline{WT}$, so _$\overline{TY} \cong \overline{WY}$_ by __definition of midpoint____. Also, it is given that Y is the midpoint of $\overline{RX}$, so _$\overline{RY} \cong \overline{XY}$_ by _definition of midpoint_. _$\angle TYR \cong \angle WYX$ by vertical angles theorem_. Therefore by _SAS congruence postulate_, Δ TYR $\cong \Delta$ WYX so by __ CPCTC_, $\overline{TR} \cong \overline{XW}$.

After releasing students to discuss question 1d with their partners, allow students to share what they find in their discussions. This will give you an opportunity to hear student misconceptions.

For struggling readers or students who may be getting hung up on the paragraph proof, consider adjusting question 1e to include a different proof format such as a flowchart or two-column proof. This will allow you to determine if students conceptually understand the correct components of the proof and may need support on paragraph proofs, or if students may need additional support with the components of the proof themselves.

5.7.3 Guided Instruction: Proofs Using CPCTC (continued)

2. Triangles CXH and XMV are shown.



Given: $\overline{CV} \perp \overline{HM}$, $\overline{HX} \cong \overline{XV}$, $\overline{HC} \cong \overline{MV}$
Prove: $\overline{XC} \cong \overline{XM}$

a. Mark the diagram to show the given information.
Marked on diagram.

b. Which congruence theorem or postulate do you think should be used?
HL Congruence

c. Complete the two-column proof.

Statement	Reason
1. $\overline{CV} \perp \overline{HM}$	1. Given
2. $\overline{HX} \cong \overline{XV}$	2. *Given*
3. $\overline{HC} \cong \overline{MV}$	2. *Given*
4. $\angle CXH$ and $\angle MXV$ are right angles	4. *Definition of Perpendicular Lines*
5. ΔCXH and ΔMXV are right triangles	5. *Definition of right triangle*
6. $\Delta CXH \cong \Delta MXV$	6. *Hypotenuse Leg Theorem*
7. $\overline{XC} \cong \overline{XM}$	7. *CPCTC*

Prior to teaching through question 2c, consider having students do a “brain storm” of everything they know and observe about these two triangles. This can help students get all their observations out and then be able to organize them into a formal two-column proof.

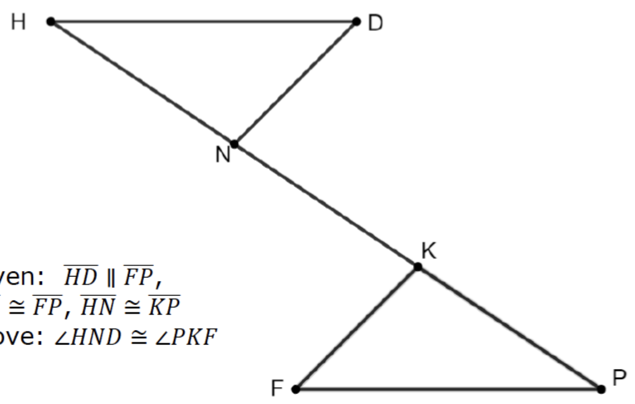
Why is SSA not something that could be used, but HL can? Think back to previous lessons.

5.7.4 Guided Practice: Proofs Using CPCTC

Component Overview: *Students will engage in a structured practice for completing triangle congruency proofs using CPCTC. For this activity, provide students with a structured opening of reviewing the image and doing the markings. Allow students to try to complete the rest on their own prior to reviewing the correct answers.*

5.7.4 Guided Practice: Proofs Using CPCTC

1. A diagram of triangles NDH and PKF with $\overline{NK}$ is shown.



Given: $\overline{HD} \parallel \overline{FP}$,
$\overline{HD} \cong \overline{FP}$, $\overline{HN} \cong \overline{KP}$
Prove: $\angle HND \cong \angle PKF$

Complete the paragraph proof.

It is given that $\overline{HD} \parallel \overline{FP}$, so by the _Alternate Interior Angles Postulate_, $\angle DHN \cong \angle FPK$. Since it is given that $\overline{HD} \cong \overline{FP}$ and $\overline{HN} \cong \overline{KP}$, $\Delta DHN \cong \Delta FPK$ by _SAS_. Therefore, $\angle HND \cong \angle PKF$ by _CPCTC_.

For this activity, consider beginning by doing a whole-class review of the given information. Allow students to determine what information is explicitly given and what additional markings they can make based on other information. Once this is completed, allow students to work on their own to complete the fill-in-the-blank components.
Think-Pair-Share
Students can work on their own to try and complete the paragraph proof and then pair up with a partner to determine the correct answers by sharing their thinking and approaches with one another.

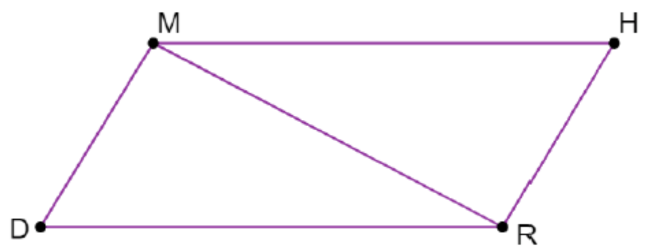
While students are working and during the review of the answer, pay close attention to students who are not grasping the content. Collect that data during this activity to be able to provide differentiated support during the next activity.

5.7.5 Your Turn!

Component Overview: *Students will demonstrate their knowledge of completing a proof using CPCTC. If students have been keeping an organized list of definitions, postulates, and theorems, allow them to reference that resource during this activity.*

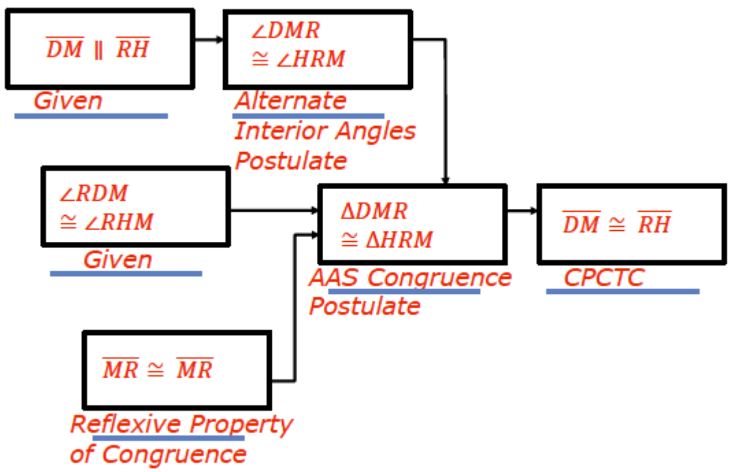
5.7.5 Your Turn!

1. Quadrilateral $MHRD$ is shown.



Given: $\overline{DM} \parallel \overline{RH}$, $\angle RDM \cong \angle RHM$
Prove: $\overline{DM} \cong \overline{RH}$

Complete the flowchart proof.



For differentiation, provide students with the opportunity to choose to complete a two-column proof or paragraph proof to demonstrate their understanding. Use their work as a starting point of remediation toward completing flowchart proofs.

5.7.6 Wrap-Up: Reflection

Component Overview: *Students will complete a self-reflection on student understanding of the content of this lesson. This activity should be done individually to provide data on individual student perception.*

5.7.6 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current learning is related to each target.

1. I understand how to use corresponding parts of congruent triangles in a proof.
   Answers will vary.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.

2. I can prove parts of triangles are congruent using triangle congruence and corresponding parts of congruent triangles.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.

Consider having students mark their reflections on each of the different components in the lesson.